SHRUTI SRIDHAR

New York, NY | shrutisridhar4@gmail.com | LinkedIn | Github | Portfolio

Skills

Languages: Python, SQL, TypeScript, Java

LLM & Retrieval: RAG pipelines, LangChain, LangGraph, LangSmith, FAISS, Pinecone, BM25, Hugging Face Transformers

ML Frameworks: PyTorch (CUDA/GPU), TensorFlow, Scikit-learn, Stable Baselines3, XGBoost, LightGBM, LoRA, PEFT, SHAP

MLOps & Backend: Airflow, FastAPI, Flask, gRPC, CI/CD, Prometheus, Grafana, PostgreSQL, Redis

Cloud & Infra: AWS (EC2, S3, SageMaker), Docker, Kubernetes, Linux, Git

Experience

Graduate Research Assistant(Co-Author), NYU Courant Institute of Mathematical Sciences — Jan 2026 - Present

Advisor: Prof. Dennis Shasha

- Built a non-Markovian epidemic forecasting pipeline across 5-20 day horizons, achieving mean RMSE 112-123 vs. 620-730 for naive baselines, a 5-6x improvement across 12 parameter settings.
- Validated forecasts against Markovian baselines on MAE, bias, and MAPE metrics using rolling-window calibration on population and contact-network inputs; currently extending evaluation across age-stratified cohorts.
- Co-authoring a manuscript on non-Markovian epidemic forecasting (preparing for submission, MDPI Algorithms, IF 2.1)

AI Software Intern, PM Accelerator - New York, NY — Sep 2025 – Dec 2025

- Scaled a high-throughput embedding pipeline (E5-Large-V2 + HNSW indexing) to serve 1M+ vectors at 5–10 ms p95 retrieval latency, enabling real-time semantic search at production scale.
- Designed end-to-end RAG evaluation workflows measuring retrieval precision@k, answer grounding rate, and latency SLAs across 4 pipeline variants, surfacing a 12% grounding accuracy gain from reranking over vanilla top-k retrieval.
- Architected a dual-retrieval inference service (gRPC primary + Python fallback) on Kubernetes achieving 99.5% uptime and cutting mean-time-to-recovery from over 2 minutes to under 10 seconds.
- Optimized embedding model fine-tuning with GPU-accelerated mixed-precision training and gradient checkpointing, reducing GPU memory usage by 30% while staying within 1% of full-precision retrieval quality.
- Deployed production monitoring dashboards tracking latency percentiles, retrieval quality scores, and failure modes, shrinking regression detection time from days to under 1 hour.

ML/AI Intern, National Institute of Ocean Technology - Chennai — Jan 2024 – May 2024

- **Co-Inventor - Design Patent: Real-Time Coral Reef Monitoring Instrument (filed 2024)**
- Trained a VGG-19 coral-health classifier on 10K+ labeled underwater images to 92% test accuracy using targeted data augmentation and a preprocessing pipeline (denoising, color correction, histogram equalization) for low-visibility conditions.
- Integrated on-device inference with LoRa wireless telemetry into the patented reef-monitoring instrument firmware, enabling continuous autonomous environmental assessment and cutting manual inspection workload by 70%.

Projects

BoxBox: F1 Pit Stop Strategy Optimizer (live link) | PPO, Gymnasium, gRPC, Flask, Prometheus, Grafana, Docker, Kubernetes

- Implemented a custom Gymnasium environment simulating F1 races lap-by-lap from 45,623 real telemetry laps across 42 Grand Prix, trained a PPO agent that gains 5.7 positions over a fixed-window baseline (P7.8 vs P13.5 mean finish) with a 1.3-stop strategy matching real F1 race patterns.
- Served the RL agent over gRPC hitting 4.67ms p95 inference (40x under the 200ms target), streamed live recommendations via Flask-SocketIO, and monitored action distribution, latency percentiles, and model confidence through a 5-panel Grafana dashboard backed by Prometheus.

CodeMentor AI: Multi-Agent Codebase Assistant (live link) | FastAPI, Next.js, RAG, LangGraph, SSE

- Orchestrated 5 LangGraph agents (planner, retrieval, analyst, mentor, memory) behind interface-driven DI across 11 FastAPI endpoints, pairing FAISS vector search with Tree-sitter AST parsing to answer codebase queries with p95 latency under 800ms over streaming SSE.
- Instrumented the full stack with Prometheus + Grafana tracking retrieval precision, latency percentiles, and utilization, then ran 3 data-driven tuning cycles that measurably improved answer relevance; containerized with Docker, deployed on Render/Vercel, and gated merges with GitHub Actions CI.

Campaign Response Prediction: ML Classification Pipeline | LightGBM, XGBoost, PostgreSQL, SHAP

- Trained LightGBM, XGBoost, and logistic regression with 5-fold stratified CV and SMOTE on a PostgreSQL-backed pipeline, hitting ROC-AUC 0.93 / F1 0.86; decile targeting showed the top 4 deciles capture 74% of responders at 40% of the contact cost, a 1.9x lift over blanket outreach.
- Structured the pipeline as a modular CLI with swappable models and data sources, layering SHAP global/local explainability and auto-exporting lift charts, segment breakdowns, and feature importance CSVs into Power BI targeting dashboards.

Education

Master of Science in Computer Engineering- New York University, New York, GPA: 3.7/4.0 — Aug 2024 – May 2026

Bachelor of Technology in CSE- Vellore Institute of Technology, India, GPA: 3.4/4 — Sep 2020 – May 2024